

CHARMAINE GUO

Wayland, MA 01778 & Evanston, IL 60201

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Authorized to work in Canada · Willing to relocate for Summer 2027 internship

EDUCATION

Northwestern University

B.S. in Materials Science and Engineering, Double Major in Data Science

June 2028

SKILLS

Computer Languages

R, Python, Java, MATLAB, HTML, CSS

Soft Skills

Project Management, Teamwork, Time-Management, Adaptability

Hard Skills

Research Methods, Experimental Design, Data Analysis, Metallography

Organizational Tools

Microsoft Excel, Google Drive

EXPERIENCE

Metallurgy Intern

General Motors (GM)

June 2026 – Present

Toledo, OH

- Optimized low-pressure carburizing recipes (15+ variants), tightening case depth control and surface hardness.
- Performed metallography in the GM materials lab on treated gear samples, including puck preparation, polishing, etching, and microscopy to characterize grain size, and martensite and retained austenite vs. spec.
- Analyzed 70+ gears in Excel, linking load position to hardness, case depth, microstructure variation.

Student Tech Aide

The Garage

September 2025 – Present

Evanston, IL

- Manage makerspace tools and inventory: 3D printers, CNCs, laser/vinyl cutters, sewing, printers, woodworking.
- Facilitate resident fabrication by translating complex project requirements into technical workflows, providing expert guidance on additive and subtractive manufacturing techniques and safety.

Undergraduate Researcher — Schuh Group

Northwestern University

Jan 2026 – March 2026

Evanston, IL

- Synthesized zircon (ZrSiO₄) coatings on micron-scale sand particles using the Pechini method, leveraging polymerizable precursors to ensure precise stoichiometric control and chemical homogeneity.
- Refined sol–gel chemistry to stabilize resin precursors and improve uniform particle encapsulation.

Quality Intern

Littelfuse

June 2025 – August 2025

Beverly, MA

- Built centralized repository with Power Query, automating ETL for 300+ legacy quality reports.
- Developed multi-parameter search tool, enabling instant queries by part, package, and leadframe attributes.
- Optimized engineering report search time from several hours to under one minute.

Undergraduate Researcher — SPIn Lab

Northwestern University

October 2024 – June 2025

Evanston, IL

- Fabricated cellulose–acrylic films and characterized UV-cured structural color without dyes.
- Used MATLAB to link cellulose concentration, crystal arrangement, and resulting structural color response.

PROJECTS

Purrfect Care

Python, PyQt5, Gemini API

Desktop (WildHacks)

March 2025

- Built a PyQt5 desktop app with wellness reminders, to-do list, and animated notifications.
- Integrated Gemini API for contextual chat with a virtual assistant cat.

Skynet System

Onshape, Arduino IDE, ESP32

Team Project (Design Thinking & Communication)

January - March 2025

- Designed solutions for remote zoo access, with 3d printed prototypes including an interactive online tour prototype built with web technologies and a Hexbug like design.
- Final deliverable: a zipline-mounted camera system enabling live video feed controlled remotely by users.

EXTRACURRICULAR ACTIVITIES

Low Voltage Team

Northwestern Formula SAE (NUFSAE)

October 2024 – Present

- Designed Power Distribution Module PCB in Altium, owning full schematic capture and layout.
- Writing wiring harness specs: connector selection, pinout mapping, and signal/safety routing.
- Leading technical onboarding for new members by delegating design tasks and conducting training sessions on RapidHarness and industry-standard documentation.